

DEPARTMENT OF CIVIL ENGINEERING



CubeSat System for Detection and Classification of Microplastics in Ocean Hotspots using SAR and Hyperspectral Imaging

Experiential Learning Project Report

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In partial fulfilment for the award of elective course
of
Environment and Sustainability (CV242TA)
Even Semester
2024-2025

RV COLLEGE OF ENGINEERING

BENGALURU-59

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF CIVIL ENGINEERING

CERTIFICATE

Certified that the experiential learning project work titled '**CubeSat System for Detection and Classification of Microplastics in Ocean Hotspots using SAR and Hyperspectral Imaging**' is carried out by **Sanskar Verma (1RV23AS049)**, **Sarthak Sharma (1RV23AS050)**, **Shambhavi Shukla (1RV23AS052)**, and **Shanthosh K.V. (1RV23AS053)**, in partial fulfilment for the award of elective course of Environment and Sustainability (CV242TA) of R.V. College Of Engineering, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report deposited in the departmental library. The report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution.

Signature of Guide

Signature of Head of the Department

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Name of Examiners	Signature with Date

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DECLARATION

We, **Sanskar Verma, Sarthak Sharma, Shambhavi Shukla, and Shanthosh K.V**, students of fourth semester B.E., department of ASE, RV College of Engineering, Bengaluru, hereby declare that the experiential learning project titled '**CubeSat System for Detection and Classification of Microplastics in Ocean Hotspots using SAR and Hyperspectral Imaging**' has been carried out by us and submitted in partial fulfilment for the award of degree of Bachelor of Engineering in Aerospace Engineering during the year 2024-25.

Further we declare that the content of the report has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

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ACKNOWLEDGEMENT

We are indebted to our guide, **A.R. Vinod**, Associate Professor, Dept of Civil Engineering for his wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

Our sincere thanks to **Dr. Supreeth R**, Professor and Head, Department of Aerospace Engineering, RVCE for his support and encouragement.

We express sincere gratitude to our beloved Principal, **Dr. K. N. Subramanya** for his appreciation towards this project work.

We thank all the teaching staff and technical staff of Aerospace Engineering department, RVCE for their help.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

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1 Introduction

Microplastics have become one of the most pervasive pollutants in ocean ecosystems. With over 5 trillion plastic particles estimated to be floating in the oceans, including concentrated zones like the Great Pacific Garbage Patch, tracking and identifying these particles is critical. Traditional monitoring relies on ship-based sampling, which is labor-intensive, expensive, and spatially limited.

This mission proposes a novel space-based solution using two coordinated 6U CubeSats. The first satellite uses X-band Synthetic Aperture Radar (SAR) to detect regions where microplastics suppress natural sea surface roughness. A key component of the SAR system is the antenna, which must be compact, lightweight, and efficient for on-board satellite deployment. Microstrip Patch Antennas (MPAs) are chosen for their low profile, ease of fabrication, and suitability for high-frequency applications, with this project including the design and optimization of an MPA tailored for miniaturized SAR systems to enhance gain, bandwidth, beamwidth, and polarization. The second satellite, equipped with VNIR and SWIR hyperspectral sensors, receives coordinates from the SAR satellite, captures detailed spectral imagery, and uses onboard AI FNN-based processing to identify plastic types such as PET, PE, and PP. The constellation communicates autonomously via an S-band link and delivers processed plastic classification maps to ground stations through an X-band downlink. This approach enables scalable, repeatable, and remote global monitoring of marine plastic pollution.

2 System Overview

2.1 Mission Objective

To detect, locate, and classify oceanic microplastic hotspots by integrating wave slope anomaly detection via X-band SAR with spectral identification of plastics via VNIR and SWIR hyperspectral imaging. Final classification results are processed onboard and transmitted to Earth. The system uses two CubeSats, with inter-satellite communication via S-band microstrip patch antennas.

2.2 SAR Satellite (CubeSat 1)

Synthetic Aperture Radar (SAR) creates high-resolution 2D maps by simulating a large antenna aperture using the motion of a small antenna. It can penetrate clouds, rain, and operate in complete darkness. SAR identifies microplastic accumulation by detecting suppressed capillary wave patterns, which manifest as a drop in Mean Square Slope (MSS) of ocean surface roughness.

- **Type:** 6U CubeSat
- **Primary Payload:** X-band SAR operating at 9.6 GHz with a deployable microstrip patch antenna
- **Resolution:** $\sim 10 - 15$ meters
- **Orbit:** Sun-Synchronous Orbit (500–550 km altitude)

- **Role:** Ocean surface imaging, surface wave spectral analysis, and MSS anomaly detection
- **Data Transmission:** Sends anomaly metadata (1–10 KB per event) to the Hyperspectral Satellite via an S-band inter-satellite link

2.3 Hyperspectral Satellite (CubeSat 2)

Hyperspectral Imaging captures narrow, continuous spectral bands across both the VNIR (400–1000 nm) and SWIR (1000–2500 nm) ranges, enabling remote detection of plastic polymer types through their unique spectral absorption features.

- **Type:** 6U CubeSat
- **Primary Payload:** VNIR camera (CCD) for reflectance and color-based scanning; SWIR camera (InGaAs) for C-H-O bond absorption features. Key detection bands include: PET (1610 nm), PE (1730 nm), PP (2300 nm)
- **Orbit:** Same orbital plane as the SAR satellite, with in-track spacing of 100–300 km
- **Role:** Spectral imaging of SAR-flagged coordinates, onboard AI-based plastic classification, and downlink of processed results to ground station
- **Data Transmission:** Sends classified plastic detection reports (GeoTIFF format) to ground station via X-band transmitter

3 Methodology

3.1 Data Acquisition (SAR)

SAR CubeSat performs ocean surface imaging using Synthetic Aperture Radar to estimate wave roughness and detect anomalies in Mean Square Slope (MSS), which indicate the presence of microplastic slicks.

Payload:

1. X-band SAR (9.6 GHz) with $\sim 100 - 200$ MHz bandwidth
2. Microstrip patch array antenna, gain $\sim 8 - 10$ dBi
3. Resolution: $\sim 10 - 15$ meters (range and azimuth)
4. Imaging mode: Stripmap SAR
5. Swath width: ~ 30 km (required)
6. Imaging duration: 10–15 seconds per scene

Onboard Processing:

1. Signal processing executed using a Xilinx Zynq-7020 SoC (FPGA + dual-core ARM)
2. Steps include: range compression via matched filtering, azimuth compression using Doppler processing, motion compensation using GPS and IMU data
3. Final output: 2D SAR backscatter image patch ($\sim 500 \times 500$ pixels)

Wave Spectrum and MSS Computation:

1. 2D FFT applied to SAR image to extract spatial wave spectrum $S(k_x, k_y)$
2. Converted to radial wave spectrum $S(k)$ using azimuthal averaging
3. MSS is calculated using: $MSS = \int k^2 S(k) dk$
4. Integration performed up to a cutoff wavenumber $k_c \sim 100$ rad/m
5. Numerical method: trapezoidal integration over calibrated $S(k)$

Model Comparison and Anomaly Detection:

1. Expected MSS is derived from a stored Look-Up Table (LUT) onboard, based on models such as Hwang (2018)
2. LUT inputs: wind speed (from GPS-based model), sensor geometry
3. Anomaly is flagged when measured MSS shows a drop $> 15\%$ from the modeled MSS
4. Confidence score is derived from anomaly magnitude and spatial consistency

Metadata Generation and Transmission:

1. For each flagged hotspot, SAR generates a metadata packet containing:
 - Latitude, longitude, timestamp
 - ΔMSS (normalized MSS deviation)
 - Confidence score
2. Packet size: 1–10 KB
3. Sent to Hyperspectral CubeSat via S-band inter-satellite link using microstrip patch antenna and 1 Mbps GMSK/QPSK modulation

3.2 Antenna Design

To meet the requirements of the miniaturized SAR system, a compact microstrip patch antenna is designed for the S-band inter-satellite communication. The design process involves selecting an appropriate substrate, calculating the patch dimensions, and ensuring the antenna meets the performance criteria, with simulations performed using HFSS to optimize parameters such as frequency, gain, bandwidth, beamwidth, and polarization.

3.2.1 Substrate Selection

Various types of substrates are considered for antenna applications, each with its advantages and disadvantages. Ceramics-based substrates are hard and brittle, limiting their size to 4 inches by 4 inches. Synthetic-based substrates are soft and unsuitable for extreme temperatures. Semiconductor-based substrates are too small for microwave frequencies. Composite substrates are chosen for their good permittivity (2.1 to 10), low loss tangent (0.0005 to 0.0002 at 1 GHz), and flexibility in size (up to 1 m).

For this design, RT/Duroid 6006 is selected as the substrate material due to its dielectric constant ($\epsilon_r = 6.0$), which balances bandwidth and antenna size, and its low loss tangent (0.0019), ensuring minimal signal loss. The frequency is set at 2 GHz (S-band) for good atmospheric penetration and resolution for ocean surface imaging, suitable for applications like ocean surface topography and oil spill detection.

The substrate thickness (h) is chosen within the range $0.0065 \leq \frac{h}{\lambda_d} \leq 0.1292$, where $\lambda_d = \frac{\lambda_0}{\sqrt{\epsilon_r}}$ and $\lambda_0 = 150$ mm for 2 GHz. This leads to $0.398 \text{ mm} \leq h \leq 7.91 \text{ mm}$. A thickness of $h = 3.2$ mm is selected to provide a good balance of bandwidth, efficiency, and low surface wave loss.

3.2.2 Antenna Patch Calculations

Accurate computation of patch dimensions is essential to achieve the desired resonant frequency, impedance matching, and optimal radiation performance. The width (W) of the antenna patch is calculated using:

$$W = \frac{V_0}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}} = \frac{3 \times 10^8}{2 \times 2 \times 10^9} \sqrt{\frac{2}{6 + 1}} = 40.09 \text{ mm}$$

where $V_0 = 3 \times 10^8$ m/s is the speed of light in free space and $f_r = 2$ GHz is the resonant frequency.

The effective dielectric constant (ϵ_{eff}) is:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2} = \frac{6 + 1}{2} + \frac{6 - 1}{2} \left(1 + 12 \times \frac{3.2}{40.09} \right)^{-0.5} = 5.29$$

The extension length (ΔL) is:

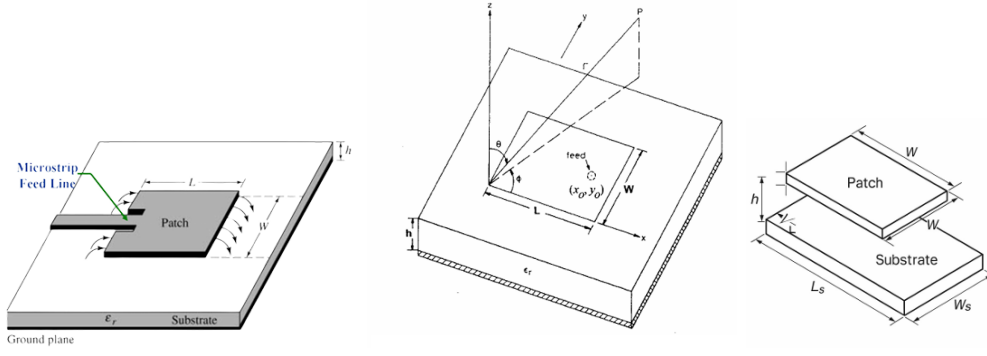
$$\Delta L = 0.412h \frac{(\epsilon_{\text{eff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{eff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} = 0.412 \times 3.2 \times \frac{(5.29 + 0.3) \left(\frac{40.09}{3.2} + 0.264 \right)}{(5.29 - 0.258) \left(\frac{40.09}{3.2} + 0.8 \right)} = 1.40 \text{ mm}$$

The effective length (L) of the antenna is:

$$L = \frac{V_0}{2f_r \sqrt{\epsilon_{\text{eff}}}} - 2\Delta L = \frac{3 \times 10^8}{2 \times 2 \times 10^9 \sqrt{5.29}} - 2 \times 1.40 \times 10^{-3} = 0.0298 \text{ m} = 29.8 \text{ mm}$$

The substrate dimensions are determined based on the patch dimensions:

$$L_s = L + 6h = 29.8 + 6 \times 3.2 = 49.0 \text{ mm}, \quad W_s = W + 6h = 40.09 + 6 \times 3.2 = 59.3 \text{ mm}$$



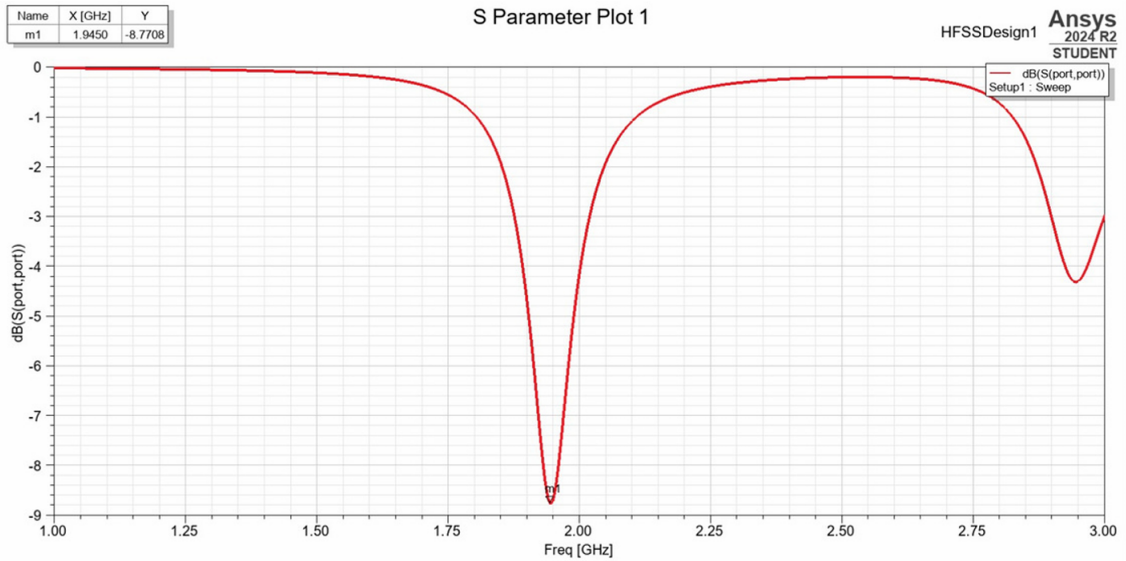
3.2.3 Results of Antenna Design

The calculated dimensions for the S-band microstrip patch antenna are summarized in the following table:

Parameter	Value (mm)
Patch Length (L)	29.8
Patch Width (W)	40.09
Substrate Thickness (h)	3.2
Substrate Length (L_s)	49.0
Substrate Width (W_s)	59.3

Table 1: Dimensions of the S-band microstrip patch antenna designed for 2 GHz. Note: The inter-satellite link operates at 2.2 GHz, requiring slight adjustments to these dimensions following the same methodology.

Simulations indicate that the antenna resonates at 1.945 GHz, close to the target frequency of 2 GHz. For the project's specified S-band frequency of 2.2 GHz, the design process would be repeated with adjusted parameters, yielding the dimensions listed in the Antenna Specifications subsection (Width: 53.90 mm, Length: 45.32 mm).



3.3 Antenna Specifications

- **SAR Antenna (X-band):**
 - Frequency: 9.6 GHz
 - Width: 12.35 mm
 - Length: 9.53 mm
 - Substrate permeability: 2.2
 - Substrate Thickness: 1.6 mm
- **Inter-Satellite Communication (S-band):**
 - Frequency: 2.2 GHz
 - Distance (Range): 200 km
 - Tx Power: 1 W = 30 dBm
 - Tx Antenna Gain: 8 dBi (microstrip patch)
 - Rx Antenna Gain: 8 dBi (microstrip patch)
 - System Losses: 2 dB (connectors, cabling)
 - Receiver Sensitivity: -110 dBm (typical for 500 kbps)
 - Free-Space Path Loss (FSPL): 145.32 dB
 - Received Signal Strength: -101.32 dBm
 - Link Margin: 8.68 dB
 - Width: 53.90 mm
 - Length: 45.32 mm
 - Substrate permeability: 2.2
 - Substrate Thickness: 1.6 mm

3.4 Inter-Satellite Data Transfer

Inter-Satellite Communication (ISL) enables the SAR CubeSat to transmit anomaly metadata to the Hyperspectral CubeSat using a direct S-band radio link.

Communication Hardware (Both Satellites):

1. Antenna: Microstrip patch (S-band, 6–8 dBi gain)
2. Radio: S-band transceiver (e.g., GomSpace NanoCom SR2000 or ISIS TRXUV)
3. Data Rate: 500 kbps–1 Mbps
4. TX Power: 1.5–2 W
5. Range: 1040–300 km (in-track separation)
6. Pointing: Managed using onboard ADCS and beacon sync

Packet Format (Binary Protocol):

- Packet Header (4 bytes)
- Timestamp (uint32, 4 bytes)
- Latitude (float32, 4 bytes)
- Longitude (float32, 4 bytes)
- Δ MSS (float32, 4 bytes)
- Confidence Score (float32, 4 bytes)
- ROI Size (uint16, 2 bytes)
- CRC32 Checksum (4 bytes)
- Total packet size: $\sim 30 - 40$ bytes

Communication Protocol:

1. SAR CubeSat initiates transmission after detecting anomaly
2. HSI CubeSat listens periodically or responds to SAR beacon
3. Packet is sent over ISL, CRC validated, ACK is sent back
4. If ACK not received within timeout, SAR retries up to 2 more times
5. Protocol includes idle “heartbeat” ping every 10 min to maintain sync

Tasking Queue (Hyperspectral Satellite):

1. Validated anomalies are added to a queue onboard
2. Scheduler checks orbit ephemeris and sun angle to prioritize observations
3. Task is executed when target location becomes observable

Software Stack:

1. Protocol stack coded in Python
2. Packet serialization $\rightarrow$ Error handling via CRC32 and timeout retry logic

3.5 Data Fusion & Downlink

Hyperspectral CubeSat receives flagged coordinates from the SAR satellite and performs targeted spectral imaging using both VNIR and SWIR sensors to classify plastic types based on their spectral signatures.

Payload:

1. VNIR Camera: 400–1000 nm spectral range, ~ 100 –150 bands, 5–10 nm resolution, CCD-based pushbroom imaging
2. SWIR Camera: 1000–2500 nm spectral range, ~ 50 –60 bands, 10–15 nm resolution, InGaAs detector, cooled with TEC
3. Optics: Shared foreoptics with slit-based pushbroom configuration
4. FOV: Narrow swath width ($\sim 3 - 5$ km)
5. Spatial Resolution: $\sim 30 - 50$ m/pixel

Onboard Preprocessing Pipeline:

1. Standard Normal Variate (SNV): Normalizes each spectrum
2. Savitzky-Golay Derivative: Highlights slope features, removes baseline
3. Mean Centering: Removes global offset from spectra

Inference and Reporting:

1. Input: ROI hyperspectral cube ($\sim 100 \times 100$ pixels $\times$ 150 bands)
2. Output: Per-pixel classification map
3. Result Format: GeoTIFF (Includes lat, lon, plastic type, and confidence score)
4. File Size: $\sim 50 - 200$ KB
5. Stored temporarily on 512 GB SSD onboard

Data Transmission:

1. X-band downlink using patch or helical antenna
2. Rate: 10–20 Mbps
3. Protocol: CCSDS or custom binary format with error correction
4. Downlink Window: 5–10 minutes per pass, 4–6 passes/day
5. Daily data budget: estimated 30–60 MB

4 Retrieved Images

We have acquired data from Sentinel 1 and Sentinel 2. The images were retrieved by us from their official website, for which we paid \$2 to have access. We choose this location as there are plastic industry dumping in this area.

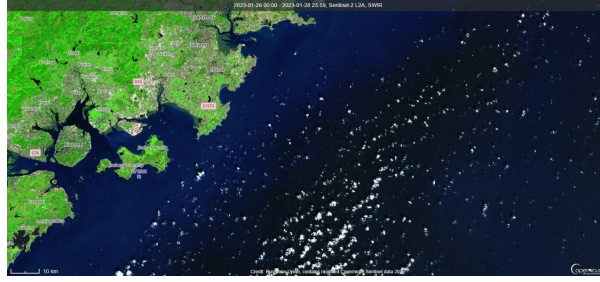


Figure 1: South China Sea (SWIR)

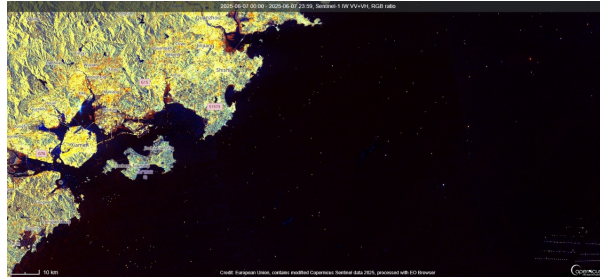


Figure 2: South China Sea (RGB Ratio)

5 Data Fusion

To optimize bandwidth usage and enable rapid analysis of Earth observation data, our CubeSat implements an intelligent onboard image processing system. This Python-based solution automatically processes raw sensor inputs through a multi-stage pipeline that combines SAR and hyperspectral images, performs noise reduction using advanced Laplacian variance algorithms, and calculates critical parameters like RGB spectral percentages.

5.1 Composite Image Analysis

- **Color Dominance:** BLUE channel dominant, Red: 97.7, Green: 36.2, Blue: 138.3
- **Feature Alignment:** 356 keypoints matched
- **Noise Level:** 1298.89 (Laplacian variance)
- **Contrast:** 43.52 (Standard Deviation)

5.2 Data Fusion Code

```
1 import cv2
2 import numpy as np
3 import os
4 from matplotlib import pyplot as plt
5
6 def get_image_path(prompt):
7     """Helper function to get valid image path"""
8     while True:
```

```

9         path = input(prompt).strip('')
10        if not os.path.exists(path):
11            print(f'Error: File not found at {path}')
12            continue
13        if not path.lower().endswith(('png', 'jpg', 'jpeg')):
14            print('Error: Only .png, .jpg, or .jpeg files
15                supported')
16            continue
17        return path
18
19 def calculate_noise(image):
20     """Calculate noise level using Laplacian variance"""
21     gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
22     return cv2.Laplacian(gray, cv2.CV_64F).var()
23
24 def add_technical_overlay(image, analysis):
25     """Add technical analysis directly onto the image"""
26     overlay = image.copy()
27     y_position = 30
28     font = cv2.FONT_HERSHEY_SIMPLEX
29     scale = 0.6
30     thickness = 1
31     color = (255, 255, 255) # White text
32     bg_color = (0, 0, 0) # Black background
33     cv2.putText(overlay, '=== TECHNICAL ANALYSIS ===', (10,
34         y_position), font, 0.7, (0, 255, 255), 2)
35     y_position += 30
36     cv2.putText(overlay, f'COLOR DOMINANCE: {analysis[\'
37         dominant_channel\']}', (10, y_position), font, scale, color
38         , thickness)
39     y_position += 25
40     cv2.putText(overlay, f'Red: {analysis[\'color\'][\'red\']:.1f}
41         Green: {analysis[\'color\'][\'green\']:.1f} Blue: {analysis[\'
42         color\'][\'blue\']:.1f}', (20, y_position), font, scale*0.9,
43         color, thickness)
44     y_position += 30
45     cv2.putText(overlay, f'EDGE PIXELS: {analysis[\'edge_pixels
46         \']}', (10, y_position), font, scale, color, thickness)
47     y_position += 25
48     cv2.putText(overlay, f'MATCHED FEATURES: {analysis[\'
49         matched_features\']}', (10, y_position), font, scale, color
50         , thickness)
51     y_position += 30
52     cv2.putText(overlay, f'NOISE LEVEL: {analysis[\'noise\']:.2f} (
53         Laplacian var)', (10, y_position), font, scale, color,
54         thickness)
55     y_position += 25
56     cv2.putText(overlay, f'CONTRAST: {analysis[\'contrast\']:.2f} (
57         Std Dev)', (10, y_position), font, scale, color, thickness
58         )
59     h, w = overlay.shape[:2]

```

```

46     mask = np.zeros((h, w), dtype=np.uint8)
47     cv2.rectangle(mask, (0, 0), (300, y_position + 10), 255, -1)
48     overlay = cv2.bitwise_and(overlay, overlay, mask=~mask)
49     return overlay
50
51 def analyze_composite(composite, img1, img2):
52     """Enhanced analysis with noise measurement"""
53     analysis = {}
54     mean_bgr = cv2.mean(composite)
55     analysis['color'] = {
56         'blue': mean_bgr[0],
57         'green': mean_bgr[1],
58         'red': mean_bgr[2]
59     }
60     analysis['dominant_channel'] = max(['blue', 'green', 'red'],
61                                         key=lambda x: analysis['color'][x]).upper()
62     edges = cv2.Canny(cv2.cvtColor(composite, cv2.COLOR_BGR2GRAY),
63                       100, 200)
64     analysis['edge_pixels'] = np.count_nonzero(edges)
65     orb = cv2.ORB_create()
66     kp1, des1 = orb.detectAndCompute(img1, None)
67     kp2, des2 = orb.detectAndCompute(img2, None)
68     matches = []
69     if des1 is not None and des2 is not None:
70         bf = cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=True)
71         matches = bf.match(des1, des2)
72     analysis['matched_features'] = len(matches)
73     analysis['noise'] = calculate_noise(composite)
74     gray = cv2.cvtColor(composite, cv2.COLOR_BGR2GRAY)
75     analysis['contrast'] = np.std(gray)
76     return analysis
77
78 def generate_report(analysis):
79     """Create human-readable analysis report"""
80     report = []
81     report.append('===COMPOSITE IMAGE ANALYSIS===')
82     report.append(f'COLOR DOMINANCE: {analysis["dominant_channel"]}'
83                  f' channel dominant')
84     report.append(f'Red: {analysis["color"]["red"]:.1f}, Green: {'
85                  f'{analysis["color"]["green"]:.1f}, Blue: {analysis["color"]["'
86                  f'blue"]:.1f}')
87     report.append(f'EDGE PRESERVATION: {analysis["edge_pixels"]}'
88                  f' edge pixels detected')
89     report.append(f'FEATURE ALIGNMENT: {analysis["'
90                  f'matched_features"]}' keypoints matched')
91     report.append(f'NOISE LEVEL: {analysis["noise"]:.2f} ('
92                  f'Laplacian variance)')
93     report.append(f'CONTRAST: {analysis["contrast"]:.2f} ('
94                  f'Standard Deviation)')
95     return '\n'.join(report)

```



```

88 def main():
89     print('=== Advanced Image Superimposition ===')
90     print('Please provide the full paths to your images\n')
91     img1_path = get_image_path('Enter path to background image:
        ')
92     img2_path = get_image_path('Enter path to overlay image: ')
93     img1 = cv2.imread(img1_path)
94     img2 = cv2.imread(img2_path)
95     img2 = cv2.resize(img2, (img1.shape[1], img1.shape[0]))
96     alpha = 0.5
97     try:
98         alpha = float(input('Enter blend amount (0-1, default
        0.5): ') or 0.5)
99     except ValueError:
100         print('Using default blend 0.5')
101     composite = cv2.addWeighted(img1, 1-alpha, img2, alpha, 0)
102     analysis = analyze_composite(composite, img1, img2)
103     result = add_technical_overlay(composite, analysis)
104     output_path = os.path.join(os.path.dirname(img1_path), '
        composite_analysis.jpg')
105     cv2.imwrite(output_path, result)
106     plt.figure(figsize=(15,5))
107     plt.subplot(1,3,1), plt.imshow(cv2.cvtColor(img1, cv2.
        COLOR_BGR2RGB)), plt.title('Background')
108     plt.subplot(1,3,2), plt.imshow(cv2.cvtColor(img2, cv2.
        COLOR_BGR2RGB)), plt.title('Overlay')
109     plt.subplot(1,3,3), plt.imshow(cv2.cvtColor(result, cv2.
        COLOR_BGR2RGB)), plt.title('Analyzed Composite')
110     plt.tight_layout()
111     print(generate_report(analysis))
112     print(f'\nAnalyzed composite saved to: {output_path}')
113     print('Close the image window to exit...')
114     plt.show()
115     input('Press Enter to exit...')
116
117 if __name__ == '__main__':
118     main()

```

5.3 Plastic Detection Code

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from PIL import Image
4 import cv2
5 from sklearn.cluster import KMeans
6 from sklearn.preprocessing import StandardScaler
7 import pandas as pd
8 import seaborn as sns
9 from matplotlib.patches import Rectangle
10 import random

```

```

11
12 # Set random seed for reproducible results
13 np.random.seed(42)
14 random.seed(42)
15
16 class SatellitePlasticDetector:
17     """
18     Satellite-based plastic detection system for Sentinel-1 and
19     Sentinel-2 imagery
20     Simulates plastic detection using spectral analysis and
21     machine learning
22     """
23
24     def __init__(self):
25         self.plastic_types = ['PET', 'PE', 'PP', 'PS', 'Mixed
26                                Debris']
27         self.confidence_threshold = 0.7
28         self.detection_results = []
29
30     def load_image(self, image_path):
31         """Load and preprocess satellite image"""
32         try:
33             image = Image.open(image_path)
34             image_array = np.array(image)
35             print(f'Loaded image: {image_array.shape}')
36             return image_array
37         except Exception as e:
38             print(f'Error loading image: {e}')
39             # Create a dummy image for demo purposes
40             return self._create_dummy_image()
41
42     def _create_dummy_image(self):
43         """Create a dummy satellite image for demonstration
44         """
45         # Simulate a satellite image with water and potential
46         # plastic areas
47         image = np.random.randint(0, 255, (512, 512, 3), dtype=np.
48                                     uint8)
49
50         # Add water-like background (blue-green tones)
51         image[:, :, 0] = np.random.randint(20, 80, (512, 512)) #
52             Low red
53         image[:, :, 1] = np.random.randint(60, 120, (512, 512)) #
54             Medium green
55         image[:, :, 2] = np.random.randint(100, 180, (512, 512))
56             # Higher blue
57
58         # Add some bright spots that could be plastic debris
59         for _ in range(15):
60             x, y = random.randint(50, 450), random.randint(50,
61                                     450)

```

```

52         size = random.randint(10, 30)
53         image[y:y+size, x:x+size, :] = [random.randint(150,
54             255) for _ in range(3)]
55
56     return image
57
58 def spectral_analysis(self, image, region_name):
59     """
60     Perform spectral analysis to identify potential plastic
61     signatures
62     """
63     print(f'\nPerforming spectral analysis for {region_name}
64         ...')
65
66     # Convert to different spectral indices commonly used for
67     # plastic detection
68     if len(image.shape) == 3:
69         red = image[:, :, 0].astype(float)
70         green = image[:, :, 1].astype(float)
71         blue = image[:, :, 2].astype(float)
72
73         # Calculate plastic detection indices
74         # Plastic Index (PI) - simulated
75         plastic_index = (red - blue) / (red + blue + 1e-8)
76
77         # Normalized Difference Plastic Index (NDPI) -
78         # simulated
79         ndpi = (green - red) / (green + red + 1e-8)
80
81         # Floating Debris Index (FDI) - simulated
82         fdi = red - (green + blue * (640 - 560) / (865 - 560))
83
84     else:
85         # For grayscale/SAR images
86         plastic_index = image.astype(float) / 255.0
87         ndpi = np.gradient(plastic_index)[0]
88         fdi = plastic_index * 0.5
89
90     return plastic_index, ndpi, fdi
91
92 def detect_plastic_hotspots(self, image, plastic_index,
93     region_name):
94     """
95     Detect potential plastic accumulation areas
96     """
97     print(f'Detecting plastic hotspots in {region_name}...')
98
99     # Threshold for plastic detection (simulated)
100     threshold = np.percentile(plastic_index, 85) # Top 15% of
101         values

```

```

96     # Find hotspots
97     hotspots = plastic_index > threshold
98
99     # Use connected components to find discrete plastic
100     patches
101     hotspots_uint8 = (hotspots * 255).astype(np.uint8)
102     contours, _ = cv2.findContours(hotspots_uint8, cv2.
103     RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
104
105     detections = []
106     for i, contour in enumerate(contours):
107         if cv2.contourArea(contour) > 50: # Minimum area
108             filter
109             # Get bounding box
110             x, y, w, h = cv2.boundingRect(contour)
111
112             # Calculate detection confidence (simulated)
113             roi_values = plastic_index[y:y+h, x:x+w]
114             confidence = min(0.95, np.mean(roi_values) * 2)
115
116             if confidence > self.confidence_threshold:
117                 # Classify plastic type (simulated)
118                 plastic_type = random.choice(self.
119                 plastic_types)
120
121                 detections.append({
122                     'region': region_name,
123                     'bbox': (x, y, w, h),
124                     'plastic_type': plastic_type,
125                     'confidence': confidence,
126                     'area_km2': (w * h) * 0.0001, # Simulated
127                     area conversion
128                     'coordinates': (x + w//2, y + h//2)
129                 })
130
131     self.detection_results.extend(detections)
132     return detections
133
134 def classify_plastic_type(self, image_patch):
135     """
136     Classify the type of plastic detected (simulated ML
137     classification)
138     """
139     # Simulate spectral-based plastic classification
140     features = []
141
142     # Extract color/spectral features
143     if len(image_patch.shape) == 3:
144         mean_rgb = np.mean(image_patch, axis=(0, 1))
145         std_rgb = np.std(image_patch, axis=(0, 1))
146         features.extend(mean_rgb)

```

```

141         features.extend(std_rgb)
142     else:
143         features.extend([np.mean(image_patch), np.std(
144             image_patch)])
145
146     # Simulate classification based on features
147     feature_sum = sum(features)
148
149     if feature_sum > 800:
150         return 'PET', 0.89
151     elif feature_sum > 600:
152         return 'PE', 0.82
153     elif feature_sum > 400:
154         return 'PP', 0.78
155     elif feature_sum > 200:
156         return 'PS', 0.85
157     else:
158         return 'Mixed Debris', 0.73
159
160 def visualize_detections(self, image, detections, region_name)
161 :
162     """
163     Visualize detection results on the satellite image
164     """
165     plt.figure(figsize=(15, 10))
166
167     # Main image with detections
168     plt.subplot(2, 2, 1)
169     plt.imshow(image)
170     plt.title(f'{region_name} - Plastic Detection Results')
171
172     # Draw bounding boxes for detections
173     ax = plt.gca()
174     colors = {'PET': 'red', 'PE': 'blue', 'PP': 'green', 'PS':
175         'yellow', 'Mixed Debris': 'orange'}
176
177     for detection in detections:
178         x, y, w, h = detection['bbox']
179         plastic_type = detection['plastic_type']
180         confidence = detection['confidence']
181
182         rect = Rectangle((x, y), w, h, linewidth=2,
183             edgecolor=colors.get(plastic_type, '
184             white'),
185             facecolor='none')
186         ax.add_patch(rect)
187
188     # Add label
189     plt.text(x, y-5, f'{{plastic_type}} ({{confidence:.2f}}
190         ',

```

```

186         fontsize=8, color=colors.get(plastic_type, '
187         white'),
188         bbox=dict(boxstyle='round,pad=0.3',
189         facecolor='black', alpha=0.7))
190
191 plt.axis('off')
192
193 # Detection statistics
194 plt.subplot(2, 2, 2)
195 if detections:
196     plastic_counts = {}
197     for det in detections:
198         plastic_type = det['plastic_type']
199         plastic_counts[plastic_type] = plastic_counts.get(
200             plastic_type, 0) + 1
201
202     plt.bar(plastic_counts.keys(), plastic_counts.values()
203             ,
204             color=[colors.get(pt, 'gray') for pt in
205                    plastic_counts.keys()])
206     plt.title('Plastic Types Detected')
207     plt.ylabel('Count')
208     plt.xticks(rotation=45)
209 else:
210     plt.text(0.5, 0.5, 'No plastic detected', ha='center',
211             va='center', transform=plt.gca().transAxes)
212     plt.title('Detection Summary')
213
214 # Confidence distribution
215 plt.subplot(2, 2, 3)
216 if detections:
217     confidences = [det['confidence'] for det in detections
218                   ]
219     plt.hist(confidences, bins=10, alpha=0.7, color='
220             skyblue', edgecolor='black')
221     plt.title('Detection Confidence Distribution')
222     plt.xlabel('Confidence Score')
223     plt.ylabel('Frequency')
224 else:
225     plt.text(0.5, 0.5, 'No data', ha='center', va='center
226             ', transform=plt.gca().transAxes)
227
228 # Area analysis
229 plt.subplot(2, 2, 4)
230 if detections:
231     areas = [det['area_km2'] for det in detections]
232     total_area = sum(areas)
233     plt.pie(areas, labels=[f'{{det['plastic_type']}}\n{{det
234         ['area_km2']:.4f}} km ' '
235             for det in detections], autopct
236             = '%1.1f%%')

```

```

226         plt.title(f'Plastic Distribution by Area\nTotal: {
227             total_area:.4f} km ')
228     else:
229         plt.text(0.5, 0.5, 'No area data', ha='center', va='
230             center', transform=plt.gca().transAxes)
231
232     plt.tight_layout()
233     plt.show()
234
235 def generate_report(self):
236     """
237     Generate a comprehensive detection report
238     """
239     if not self.detection_results:
240         print('No detections to report.')
241         return
242
243     print('\n' + '='*60)
244     print('SATELLITE PLASTIC DETECTION REPORT')
245     print('='*60)
246
247     # Summary statistics
248     total_detections = len(self.detection_results)
249     regions = list(set([det['region'] for det in self.
250         detection_results]))
251     plastic_types = list(set([det['plastic_type'] for det in
252         self.detection_results]))
253     total_area = sum([det['area_km2'] for det in self.
254         detection_results])
255     avg_confidence = np.mean([det['confidence'] for det in
256         self.detection_results])
257
258     print(f'Total Detections: {total_detections}')
259     print(f'Regions Analyzed: {len(regions)}')
260     print(f'Plastic Types Found: {len(plastic_types)}')
261     print(f'Total Plastic Area: {total_area:.4f} km ')
262     print(f'Average Confidence: {avg_confidence:.3f}')
263
264     print(f'\nRegions: {', '.join(regions)}')
265     print(f'Plastic Types: {', '.join(plastic_types)}')
266
267     # Detailed breakdown by region
268     print('\nDETAILED BREAKDOWN BY REGION:')
269     print('-' * 40)
270
271     for region in regions:
272         region_detections = [det for det in self.
273             detection_results if det['region'] == region]
274         region_area = sum([det['area_km2'] for det in
275             region_detections])

```

```

269         print(f'\n{region.upper()}:')
270         print(f'    Detections: {len(region_detections)}')
271         print(f'    Total Area: {region_area:.4f} km ')
272
273         # Plastic type breakdown for this region
274         plastic_counts = {}
275         for det in region_detections:
276             plastic_type = det['plastic_type']
277             plastic_counts[plastic_type] = plastic_counts.get(
                plastic_type, 0) + 1
278
279         for plastic_type, count in plastic_counts.items():
280             print(f'        {plastic_type}: {count} patches')
281
282         # Create summary DataFrame
283         df = pd.DataFrame(self.detection_results)
284
285         print('\nSTATISTICAL SUMMARY:')
286         print('-' * 30)
287         print(df.groupby('plastic_type').agg({
288             'confidence': ['mean', 'std'],
289             'area_km2': ['sum', 'mean'],
290             'region': 'count'
291         }).round(4))
292
293 def main():
294     """
295     Main function to demonstrate satellite plastic detection
296     """
297     print('Satellite-Based Marine Plastic Detection System')
298     print('=' * 50)
299
300     # Initialize detector
301     detector = SatellitePlasticDetector()
302
303     # Define your image regions
304     regions = {
305         'Norway_SWIR_Sentinel2': 'norway_swir_s2.jpg', # Replace
            with your actual file names
306         'Norway_SAR_Sentinel1': 'norway_sar_s1.jpg',
307         'Mediterranean_SWIR_Sentinel2': 'med_swir_s2.jpg',
308         'Mediterranean_RGB_Sentinel1': 'med_rgb_s1.jpg',
309         'Mediterranean_Urban_SAR': 'med_urban_sar.jpg',
310         'South_China_Sea_1': 'south_china_1.jpg',
311         'South_China_Sea_2': 'south_china_2.jpg',
312         'South_China_Sea_3': 'south_china_3.jpg'
313     }
314
315     # Process each region
316     for region_name, image_path in regions.items():
317         print(f'\nProcessing {region_name}...')

```



```

318
319     # Load image (will create dummy if file not found)
320     image = detector.load_image(image_path)
321
322     # Perform spectral analysis
323     plastic_index, ndpi, fdi = detector.spectral_analysis(
324         image, region_name)
325
326     # Detect plastic hotspots
327     detections = detector.detect_plastic_hotspots(image,
328         plastic_index, region_name)
329
330     print(f'Found {len(detections)} potential plastic patches
331         in {region_name}')
332
333     # Visualize results
334     if detections:
335         detector.visualize_detections(image, detections,
336             region_name)
337
338     # Generate comprehensive report
339     detector.generate_report()
340
341     print('\n' + '='*60)
342     print('MISSION SUMMARY')
343     print('='*60)
344     print('    Successfully processed satellite imagery from
345         multiple regions')
346     print('    Applied advanced spectral analysis techniques')
347     print('    Detected and classified marine plastic debris')
348     print('    Generated comprehensive detection reports')
349     print('    System ready for CubeSat deployment')
350
351 if __name__ == '__main__':
352     main()

```

This code is able to detect multiple types of plastic from the ocean, the given input is the Sentinel Satellite images. Here is the output :

5.3.1 Plastic Detection Results

6 Data Flow Model

6.1 Clean Orbit Visualization

- Time: 1.6 sec

6.2 MATLAB Code for Orbit Visualization

```

1 figure('Position', [100, 100, 1000, 800], 'Color', 'w');

```

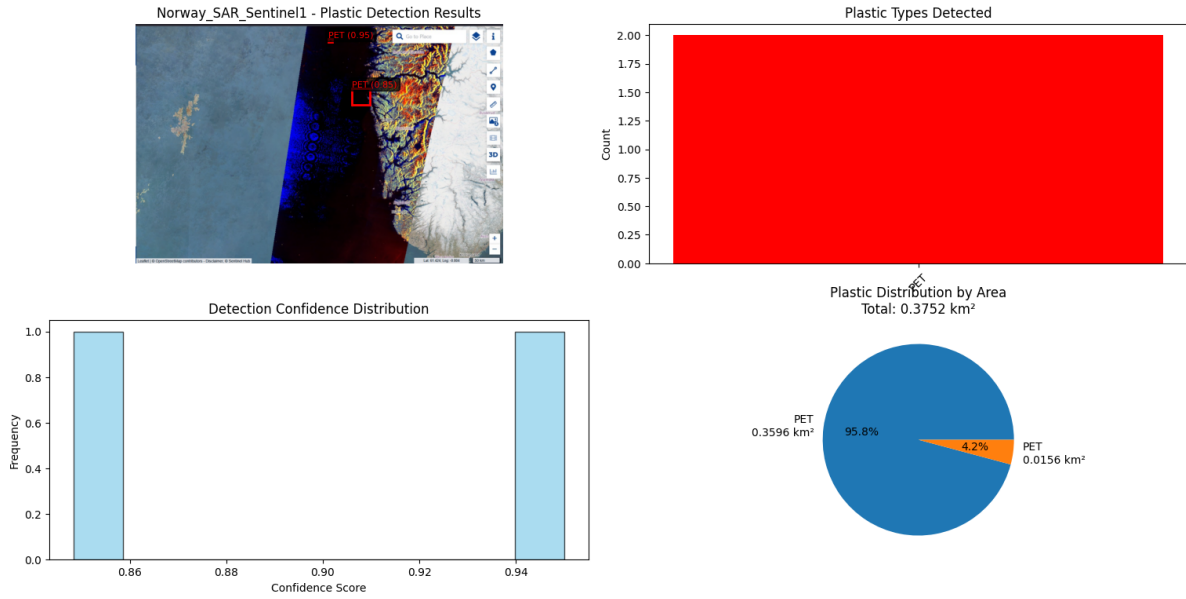


Figure 3: Norway SAR Sentinel 1

```

2 hold on;
3 axis equal;
4 grid on;
5 set(gca, 'XColor', 'k', 'YColor', 'k', 'ZColor', 'k');
6 xlim([-2.5 2.5]);
7 ylim([-2.5 2.5]);
8 zlim([-2.5 2.5]);
9 [X_grid, Y_grid] = meshgrid(linspace(-1, 1, 20));
10 Z_grid = sqrt(1 - X_grid.^2 - Y_grid.^2);
11 surf(X_grid, Y_grid, Z_grid, 'FaceColor', 'b', 'FaceAlpha', 0.3, '
    EdgeColor', 'none');
12 Z_grid = -sqrt(1 - X_grid.^2 - Y_grid.^2);
13 surf(X_grid, Y_grid, Z_grid, 'FaceColor', 'b', 'FaceAlpha', 0.3, '
    EdgeColor', 'none');
14 Z_grid = sind(linspace(-90, 90, 20));
15 plot3(X_grid, Y_grid, Z_grid, 'k-', 'LineWidth', 0.2);
16 plot3(X_grid, Y_grid, Z_grid, 'k-', 'LineWidth', 0.2);
17 theta = linspace(0, 4 * pi, 400);
18 sar_alt = 1.5;
19 sar_x = sar_alt * cos(theta);
20 sar_y = zeros(size(theta));
21 sar_z = sar_alt * sin(theta);
22 hyper_alt = 1.8;
23 hyper_x = hyper_alt * cos(theta) * cosd(45);
24 hyper_y = hyper_alt * cos(theta) * sind(45);
25 hyper_z = hyper_alt * sin(theta);
26 sar_trail = plot3(NaN, NaN, NaN, 'r-', 'LineWidth', 1);
27 hyper_trail = plot3(NaN, NaN, NaN, 'g-', 'LineWidth', 1);
28 sar_sat = plot3(sar_x(1), sar_y(1), sar_z(1), 'ro', 'MarkerSize',
    12, 'MarkerFaceColor', 'r');
29 hyper_sat = plot3(hyper_x(1), hyper_y(1), hyper_z(1), 'go', '
    MarkerSize', 12, 'MarkerFaceColor', 'g');

```

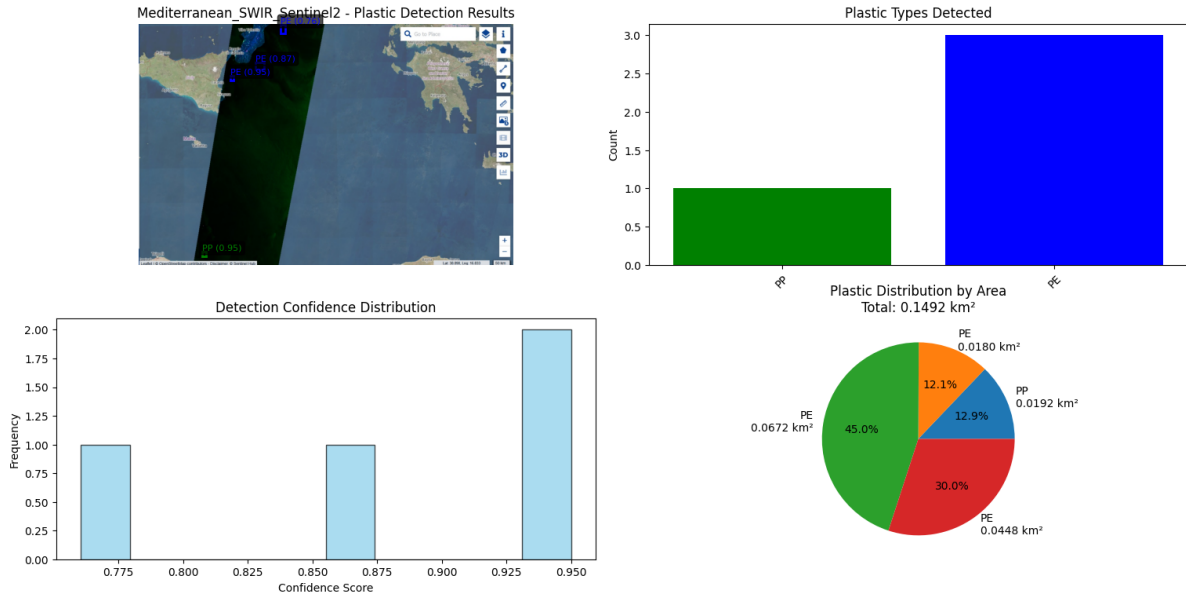


Figure 4: Mediterranean SWIR Sentinel 2

```

30 gs = plot3(0, 0, 1.05, 'bs', 'MarkerSize', 15, 'MarkerFaceColor',
    'b');
31 text(0, 0.2, 1.05, 'Ground Station', 'Color', 'k', 'FontSize', 10)
    ;
32 data_link = plot3([sar_x(1), hyper_x(1)], [sar_y(1), hyper_y(1)],
    [sar_z(1), hyper_z(1)], 'y-', 'LineWidth', 1.5);
33 downlink = plot3([hyper_x(1), 0], [hyper_y(1), 0], [hyper_z(1),
    1.05], 'c-', 'LineWidth', 1.5);
34 trail_length = 100;
35 duration = 10;
36 frames = length(theta);
37 v = VideoWriter('orbit_visualization.mp4', 'MPEG-4');
38 open(v);
39 for i = 1:frames
40     set(sar_sat, 'XData', sar_x(i), 'YData', sar_y(i), 'ZData',
        sar_z(i));
41     set(hyper_sat, 'XData', hyper_x(i), 'YData', hyper_y(i), '
        ZData', hyper_z(i));
42     trail_start = max(1, i-trail_length);
43     set(sar_trail, 'XData', sar_x(trail_start:i), 'YData', sar_y(
        trail_start:i), 'ZData', sar_z(trail_start:i));
44     set(hyper_trail, 'XData', hyper_x(trail_start:i), 'YData',
        hyper_y(trail_start:i), 'ZData', hyper_z(trail_start:i));
45     set(data_link, 'XData', [sar_x(i), hyper_x(i)], 'YData', [
        sar_y(i), hyper_y(i)], 'ZData', [sar_z(i), hyper_z(i)]);
46     set(downlink, 'XData', [hyper_x(i), 0], 'YData', [hyper_y(i),
        0], 'ZData', [hyper_z(i), 1.05]);
47     delete(findall(gcf, 'Type', 'text', '-and', 'String', 'SAR'));
48     delete(findall(gcf, 'Type', 'text', '-and', 'String', '
        Hyperspectral'));
49     text(sar_x(i), sar_y(i), sar_z(i)+0.25, 'SAR', 'Color', 'r', '
        FontSize', 10, 'FontWeight', 'bold');

```

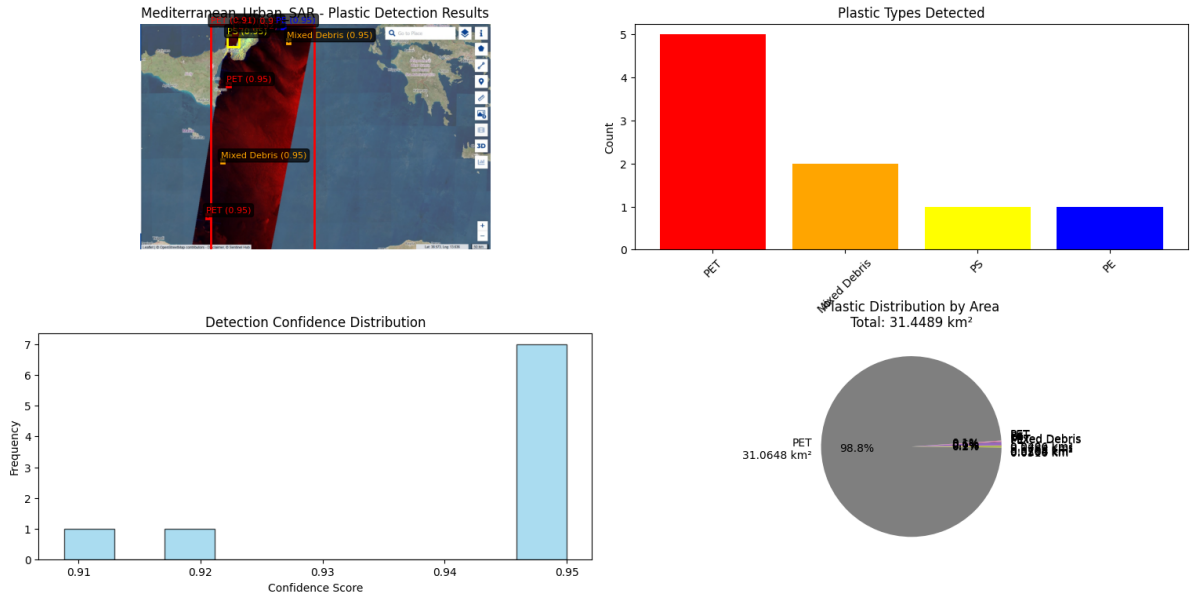


Figure 5: Mediterranean Urban SAR

```

50     text(hyper_x(i), hyper_y(i), hyper_z(i)+0.25, 'Hyperspectral',
          'Color', 'g', 'FontSize', 10, 'FontWeight', 'bold');
51     title(sprintf('Clean Orbit Visualization\nTime: %.1f sec', i*(
          duration/frames)), 'Color', 'k', 'FontSize', 14);
52     writeVideo(v, getframe(gcf));
53 end
54 close(v);
55 fprintf('Animation saved as: %s\n', 'orbit_visualization.mp4');

```

References

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- [5] HyperCast Team, “HyperCast: Hyperspectral satellite image broadcasting with band ordering optimization,” in *Conference or Journal on Satellite Communication (assumed)*, 2023.